A Numerical Certificate for the I.I.D. Lower Bound

Correctness of the original FRLP evaluator

Abstract

This document supplies the implementation correctness arguments for the fixed-coefficient factor-revealing LP in Section 3.3 and Appendix A.1 of *Mean-only Prophet Inequalities*. It proves that the rate-tree pruning rules, numerical table adjustments, interpolation, and explicit arithmetic error allowances preserve all original LP inequalities. The exact rational data and the frozen evaluator accompany the proof. The result is a feasible certificate; no optimality of the LP solution or of the chosen threshold is asserted.

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1 Certificate, scope, and notation

The manuscript reference is *Mean-only Prophet Inequalities*. References such as Lemma A.2 and Section 3.3 refer to that manuscript; numbered results in this document have their own numbering. The supplied rational threshold has $N = 40$ pieces. Its initial value and the certified objective are

$$\theta(0) = \frac{57507834299659}{77507834299659}, \qquad \gamma = \frac{815733368877}{1099511627776}.$$

The exact target comparison is

$$10000 \cdot 815733368877 - 7419 \cdot 1099511627776 = 56922299856 > 0.$$

The proof below concerns feasibility of the original finite LP. The paper's analytical transfer theorem gives the subsequent all-finite-n consequence. The historical search for the threshold is not an input to the correctness argument.

We retain the paper's symbols $b, \theta, c, p, u, \Delta, t_k, r_j$ and $v_{k,i,j}$ with their original meanings. The integer arrays in the program represent the exact rational values $v_{k,i,j}$ on a scale of 2^{40}. The state knots have denominator 2^{30}. The only interpolant used below, $\widetilde{v}_k$, is the ordinary piecewise-affine interpolation of the stored layer-k table. It need not be concave. Concavity of the true function V_t, used to relate the finite LP to the reward guarantee, is supplied by the paper.

The main files fixing this particular instance are:

File inside `certificate/`	Mathematical content
`data/threshold_slopes.json`	The exact rational drop rates $b_1, \dots, b_{40}$
`data/grid1200.json`	The shared state knots $y_i = x_i$
`data/rates_refined_Q131093.json`	The finite rate endpoints; the last upper endpoint is $+\infty$
`data/coeff_N640_Q131093.h`	Binary64 inputs with proved rational enclosures
`verify.cpp`	The frozen backward evaluator
`validate_inputs.py`	Exact-rational validation of the coefficient inputs

The preflight script also checks that the text grid actually read by the evaluator is identical to the validated JSON grid, and that the two rational representations of the threshold agree.

2 The original i.i.d. FRLP and its accelerated evaluation

This section proves that the accelerated backward evaluator constructs a feasible solution of the fixed-coefficient FRLP in Section 3.3 of the paper. The mathematical inputs are Lemmas 3.3, A.2, and A.3 of the paper. The additional arguments below justify the evaluation of that same finite LP: they do not introduce stronger reward inequalities or replace any of its rows. Arithmetic lower enclosures are established separately below. In particular, every comparison used for pruning must use such an enclosure, rather than a rounded estimate without a proved error bound.

2.1 Fixed data and literal rows

The certificate fixes $N = 40$ positive nondecreasing drop rates, $N_t = 640$ time intervals, $N_y = 1200$ state subdivisions, and $N_q = 131093$ rate intervals. Write $t_k = k/N_t$ and $\Delta = 1/N_t$. The state knots satisfy

$$0 = y_0 = x_0 < y_1 = x_1 < \cdots < y_{N_y} = x_{N_y} = 1,$$

and the rate endpoints satisfy

$$0 = r_0 < r_1 < \cdots < r_{N_q - 1} < r_{N_q} = +\infty.$$

All finite endpoints and drop rates are fixed rational inputs. On each state rectangle we use the diagonal from its lower-left to its upper-right vertex, retaining only the part of the grid in $0 \le x \le y \le 1$.

Fix a time $t = t_k$, a state $0 \le x < y \le 1$, and the constant drop rate $b > 0$ on $[t, t + \Delta]$. As in Appendix A.1, let

$$c = \max_{0 \le s \le 1} \{b(s) + \theta(s)\} - 1 > 0, \qquad p = cx + y(1 - \theta(t) - b), \qquad u = b(1 - y + ty).$$

For finite $r \geq 0$, put

$$E(r) = e^{-\Delta r}, \qquad I_r = \int_0^{\Delta} e^{-rs}\, ds = \begin{cases} (1 - E(r))/r, & r > 0, \\ \Delta, & r = 0. \end{cases}$$

The successor quantities in Lemma A.2, viewed as functions of a finite lower endpoint r, are

$$\varphi(r) = 1 - y + y\big(t + I_r + E(r)(1 - t - \Delta)\big), \tag{1}$$

$$y(r) = \frac{yE(r)}{\varphi(r)}, \tag{2}$$

$$x(r) = \min\left\{ y(r), \frac{E(r)\big(cx + yb\Delta(r + e^{-r} - 1)\big)}{c\varphi(r)} \right\}. \tag{3}$$

Here $\varphi(r) > 0$ and $0 \leq x(r) \leq y(r) \leq 1$.

For finite r, define the parameters of Lemma A.3 by

$$\begin{aligned} C(r) &= \min\{p\Delta, pI_r\}, \\ \alpha(r) &= ybE(r)\Delta^2 + (-p)^+ I_r \Delta, \\ \beta(r) &= \Delta(u + ybI_r), \qquad z(r) = \beta(r)e^{-r}. \end{aligned}$$

Both $\alpha(r)$ and $\beta(r)$ are positive at a non-diagonal state. The maximum of the two unrestricted bounds in Lemma A.3 is

$$\mathcal{H}(\alpha, \beta) = \begin{cases} \beta - \dfrac{\beta^2}{4\alpha}, & \beta \leq \alpha, \\ \alpha\left(\dfrac{3}{4} + \dfrac{1}{2}\log\dfrac{\beta}{\alpha}\right), & \beta > \alpha. \end{cases} \tag{4}$$

Indeed, with $v = \beta/\alpha$, the difference between the quadratic and logarithmic expressions, divided by α, is

$$v - \frac{v^2}{4} - \frac{3}{4} - \frac{1}{2}\log v.$$

It vanishes at $v = 1$ and has derivative $-(v-1)^2/(2v) \leq 0$. This proves the stated choice of branch. When $z(r) > \alpha(r)/2$, write

$$J(r) = z(r) + \frac{\alpha(r)r}{2} - \frac{\alpha(r)^2}{24(z(r) - \alpha(r)/2)}.$$

For this fixed time and state, suppress the indices k, i, j in the cost coefficient. For the literal rate interval $[r_{h-1}, r_h]$, let a_h be the maximum of

$$\min\{p\Delta, 0\}, \qquad C(r_h) + \mathcal{H}(\alpha(r_h), \beta(r_h)), \qquad C(r_h) + J(r_h),$$

where the last term is included only when its guard holds, and the latter two terms are omitted when $r_h = +\infty$. Thus $a_{N_q} = \min\{p\Delta, 0\}$. These are exactly the constants specified at the end of Appendix A.1, before numerical lower enclosures are taken.

Let $v_{k+1,i,j}$ be the already fixed next-layer table, and let $\widetilde{v}_{k+1}(y, x)$ denote its ordinary continuous piecewise-affine interpolant on the stated triangles. The right-hand side of the literal row indexed by h is

$$\mathcal{R}_h = a_h + \varphi(r_{h-1})\, \widetilde{v}_{k+1}(y(r_{h-1}), x(r_{h-1})). \tag{5}$$

The current table entry must satisfy $v_{k,i,j} \leq \mathcal{R}_h$ for every h. The code's zero-based leaf index $h - 1$ uses `A(h-1)` for the continuation and `B(h)` for the running cost (`verify.cpp`, lines 63–96 and 134).

The evaluator may use a lower bound on $\mathcal{R}_h$ that is strictly smaller than the literal expression. In particular, omitting an applicable lower bound from the maximum defining a_h, imposing a stronger guard before using J, or subtracting a nonnegative allowance cannot invalidate a candidate that is required to lie below the literal row. None of these operations changes the target LP.

2.2 Lowering the numerical table

The monotonicity of the true functions V_t does not by itself imply monotonicity of an arbitrary numerical table. We enforce the property needed below by lowering the newly computed layer, in the following order:

$$\begin{aligned} &i = 1, \dots, N_y, \quad j = i-1, i-2, \dots, 0 : && v_{k,i,j} \leftarrow \min\{v_{k,i,j}, v_{k,i,j+1}\}; \\ &i = 1, \dots, N_y, \quad j = 0, \dots, i-1 : && v_{k,i,j} \leftarrow \min\{v_{k,i,j}, v_{k,i-1,j}\}. \end{aligned}$$

Diagonal entries remain zero. These are the two passes in `lower_monotone` (lines 158–160).

Lemma 2.1 (Feasibility and monotonicity after lowering)**.** *Suppose the next layer is fixed, the current layer is nonpositive with zero diagonal, and every current literal row is satisfied. The two passes preserve those properties and yield*

$$v_{k,i,j} \le v_{k,i,j+1}, \qquad v_{k,i,j} \le v_{k,i-1,j} \qquad (0 \le j < i \le N_y).$$

Its interpolant is nonpositive, zero on $x = y$, nondecreasing in x, and nonincreasing in y throughout the triangular state domain.

Proof. Each row constraint for layer k has a current entry only on its left-hand side; its right-hand side uses the already fixed layer $k+1$. Decreasing a current entry therefore preserves every such constraint. The passes leave the diagonal unchanged and only decrease nonpositive entries. They are applied before any earlier layer is constructed, so there are no already constructed constraints from layer $k-1$ to preserve.

The descending order of j in the first pass makes each row nondecreasing in j. In the second pass, row $i-1$ has already been completed. The new row i is the pointwise minimum of its previous nondecreasing row and the completed nondecreasing row $i-1$, with the latter extended by a final zero. Hence it remains nondecreasing. Each column inequality is imposed against the completed preceding row and cannot be changed by a later update. This proves both displayed inequalities.

For the interpolation claim, consider a rectangle with coordinate widths $d_y = y_{i+1} - y_i > 0$ and $d_x = x_{j+1} - x_j > 0$. On its lower triangle with vertices (y_i, x_j), (y_{i+1}, x_j), (y_{i+1}, x_{j+1}), the two partial derivatives are

$$\partial_x \widetilde{v}_k = \frac{v_{k,i+1,j+1} - v_{k,i+1,j}}{d_x} \ge 0, \qquad \partial_y \widetilde{v}_k = \frac{v_{k,i+1,j} - v_{k,i,j}}{d_y} \le 0.$$

On the upper triangle they are

$$\partial_x \widetilde{v}_k = \frac{v_{k,i,j+1} - v_{k,i,j}}{d_x} \ge 0, \qquad \partial_y \widetilde{v}_k = \frac{v_{k,i+1,j+1} - v_{k,i,j+1}}{d_y} \le 0.$$

Only the lower triangle is present when $i = j$. Continuity across edges extends these monotonicities along any horizontal or vertical segment in the domain. Every interpolated value is a convex combination of nonpositive entries, and interpolation along the diagonal uses only zero diagonal entries. □

2.3 A continuation bound for an entire rate subtree

A tree node consists of consecutive literal intervals. Write r_- for its first lower endpoint and r_+ for its last upper endpoint; r_- is finite and r_+ may be infinite. Every lower endpoint occurring in the node lies in $[r_-, r_+]$. Define

$$\rho(r) = \frac{x}{y} + \frac{b\Delta}{c}(r + e^{-r} - 1), \qquad x(r) = y(r)\min\{1, \rho(r)\}.$$

Lemma 2.2 (Uniform continuation bound)**.** *Suppose $\widetilde{v}_{k+1}$ has the properties in Lemma 2.1. If $\rho(r_-) \geq 1$, every continuation in the node is zero. Otherwise define*

$$y_+ = \begin{cases} y(r_+), & r_+ < +\infty, \\ 0, & r_+ = +\infty, \end{cases} \qquad x_- = y_+\rho(r_-).$$

For every finite $r \in [r_-, r_+]$,

$$\varphi(r)\widetilde{v}_{k+1}(y(r), x(r)) \geq \varphi(r_-)\widetilde{v}_{k+1}(y(r_-), x_-). \tag{6}$$

The bound remains valid if y_+ is replaced by any certified value in $[0, y_+]$, in particular by zero.

Proof. The functions $E(r)$ and I_r are nonincreasing. Since $1 - t - \Delta \geq 0$, equation (1) shows that φ is nonincreasing. Moreover,

$$\frac{\varphi(r)}{E(r)} = (1 - y)e^{\Delta r} + y\left(te^{\Delta r} + \int_0^{\Delta} e^{rs}\, ds + 1 - t - \Delta\right)$$

is nondecreasing, so $y(r) = y/(\varphi(r)/E(r))$ is nonincreasing. Finally $\rho'(r) = b\Delta(1 - e^{-r})/c \geq 0$. If $\rho(r_-) \geq 1$, it follows that $x(r) = y(r)$, where the interpolant is zero.

Otherwise $0 \leq \rho(r_-) < 1$. For every finite r under consideration, $y(r) \geq y_+$ and $\min\{1, \rho(r)\} \geq \rho(r_-)$; hence $x(r) \geq x_-$. Also $y(r) \leq y(r_-)$, and $0 \leq x_- \leq y_+ \leq y(r_-)$, so the comparison state belongs to the domain. Coordinate monotonicity gives

$$\widetilde{v}_{k+1}(y(r), x(r)) \geq \widetilde{v}_{k+1}(y(r_-), x_-).$$

Multiplying by $\varphi(r) > 0$, and then using $\varphi(r) \leq \varphi(r_-)$ and the nonpositivity of the right-hand interpolated value, proves (6). Replacing y_+ by a smaller nonnegative value only decreases x_- and therefore weakens this lower bound. □

This is the exact-arithmetic argument behind `rectA` (lines 98–106). The implementation uses zero for the far endpoint unless its arithmetic guard permits an accurately enclosed division. Its coordinate perturbations and multiplier rounding are justified by the arithmetic enclosures below; they are not assumptions of exact monotonicity for rounded expressions.

2.4 A running-cost bound for an entire rate subtree

Lemma 2.3 (Monotonicity of the free cost bound)**.** *For fixed t, x, y, b, the functions $C(r)$, $\alpha(r)$, and $\beta(r)$ are nonincreasing in finite $r \geq 0$. The function $\mathcal{H}(\alpha, \beta)$ is nondecreasing in each positive argument. Consequently, if $r_+ < +\infty$, then every literal cost in the node is at least*

$$C(r_+) + \mathcal{H}(\alpha(r_+), \beta(r_+)). \tag{7}$$

Every literal cost, including the unbounded leaf, is also at least $\min\{p\Delta, 0\}$.

Proof. If $p < 0$, then $C(r) = p\Delta$ because $I_r \le \Delta$; if $p \ge 0$, then $C(r) = pI_r$. Thus C is nonincreasing. The formulas for α and β are nonnegative linear combinations of $E(r)$, I_r, and constants, so the same conclusion holds for them. For $\beta \le \alpha$,

$$\partial_\alpha \mathcal{H} = \frac{\beta^2}{4\alpha^2} > 0, \qquad \partial_\beta \mathcal{H} = 1 - \frac{\beta}{2\alpha} \ge \frac{1}{2}.$$

For $\beta \ge \alpha$,

$$\partial_\alpha \mathcal{H} = \frac{1}{4} + \frac{1}{2}\log\frac{\beta}{\alpha} > 0, \qquad \partial_\beta \mathcal{H} = \frac{\alpha}{2\beta} > 0.$$

The formulas and their first derivatives agree at $\beta = \alpha$, which proves coordinate monotonicity on the whole positive quadrant. Each finite leaf in the node has upper endpoint at most r_+, and its literal cost includes this free bound evaluated at its own upper endpoint. The claimed inequalities follow. □

Lemma 2.4 (Uniform shifted cost bound)**.** *Suppose $r_+ < +\infty$ and $z(r_+) > \alpha(r_-)$. Then every literal cost in the node is at least*

$$C(r_+) + z(r_+) + \frac{\alpha(r_+)r_-}{2} - \frac{\alpha(r_-)^2}{24\big(z(r_+) - \alpha(r_-)/2\big)}. \tag{8}$$

Proof. Both factors in $z(r) = \beta(r)e^{-r}$ are nonnegative and nonincreasing, so z is nonincreasing. At every finite upper endpoint $r \in [r_-, r_+]$ of a descendant leaf,

$$z(r) \ge z(r_+) > \alpha(r_-) \ge \alpha(r) > \alpha(r)/2.$$

Thus its literal cost includes the shifted bound $C(r) + J(r)$. The positive terms satisfy

$$C(r) \ge C(r_+), \qquad z(r) \ge z(r_+), \qquad \alpha(r)r \ge \alpha(r_+)r_-.$$

For the subtracted term, its denominator is positive and

$$\frac{\alpha(r)^2}{z(r) - \alpha(r)/2} \le \frac{\alpha(r_-)^2}{z(r_+) - \alpha(r_-)/2}.$$

Combining these inequalities proves the result. No monotonicity of the shifted expression $J(r)$ itself is asserted or required. □

The maximum of the available bounds in the preceding two lemmas is therefore a valid running-cost lower bound for the node. For a node with $r_+ = +\infty$ we retain only $\min\{p\Delta, 0\}$. This is the argument used by `Bnode` (lines 108–124). Its stronger numerical guards, including $r_+ \le 64$, only suppress optional bounds. The implemented free bound uses a conservative logarithm evaluation in `log_lower`; no polynomial replacement of $\mathcal{H}$ is used or needed for this certificate.

2.5 Complete coverage and backward feasibility

For each node, add a certified lower enclosure of (6) to a certified lower enclosure of an applicable running-cost bound. Call the resulting quantity $\underline{\mathcal{R}}_{\text{node}}$. By the preceding lemmas,

$$\underline{\mathcal{R}}_{\text{node}} \le \mathcal{R}_h \quad \text{for every literal leaf } h \text{ in that node.} \tag{9}$$

For an individually evaluated leaf, write $\underline{\mathcal{R}}_h \le \mathcal{R}_h$ for the certified enclosure returned by its continuation and cost evaluations.

The default run subtracts 2^{-28} from each candidate. It also subtracts a nonnegative quantity of the form

$$2^{-40}\left\lceil 2^{24} d_h \right\rceil, \qquad 0 \le d_h \le y - x \le 1,$$

where the last inequalities are enforced by clamping in `allowance` (lines 126–132). Its detailed formula is only a candidate-selection policy. For feasibility, the only fact needed is that the subtracted quantity is nonnegative. In particular,

$$\underline{\mathcal{R}}_h - 2^{-28} - 2^{-40}\lceil 2^{24} d_h \rceil \le \mathcal{R}_h.$$

No assertion about a telescoping potential or approximation to an exact potential decrement is used.

Theorem 2.5 (Feasibility of the accelerated backward evaluator)**.** *Assume the arithmetic routines return the lower enclosures specified above. Starting from the zero terminal table, the evaluator's recursive search and subsequent lowering passes produce a feasible solution of the original FRLP.*

Proof. Fix a state and a previously fixed next layer. The algorithm maintains a nonpositive incumbent w, initialized by the minimum of zero and conservative candidate values. Every update takes a minimum with $\underline{\mathcal{R}}_h - 2^{-28} - 2^{-40}\lceil 2^{24} d_h \rceil$, so w can only decrease. The initial hint, the unbounded leaf, and the ternary-search proposals supply candidates only; they do not remove any leaf from the subsequent search.

A node is pruned only if $\underline{\mathcal{R}}_{\text{node}} \ge w$. At that moment, (9) implies $w \le \mathcal{R}_h$ for every leaf in that node. Later decreases of w preserve all these inequalities. If a node is not pruned and is a singleton, its leaf is evaluated and the incumbent is updated; the resulting value is at most that leaf's literal row. Otherwise the node is partitioned into two disjoint nonempty consecutive sets of leaves and both children are searched.

The root contains all N_q leaves, including the last unbounded interval. Recursive splitting strictly decreases the number of leaves, so the search terminates. Its terminal nodes form a disjoint cover of the root. Every terminal node is either pruned with its inequalities proved, or explicitly evaluated. Consequently the final incumbent satisfies $w \le \mathcal{R}_h$ for every h. This argument also covers an immediate pruning of the root. The source's coverage counter, which adds the number of leaves at a pruned node and one at an evaluated singleton, checks this complete cover; the counter alone is not a substitute for (9).

Apply this search independently to each non-diagonal state. Its outputs are nonpositive and satisfy every literal row against the fixed next layer; the diagonal is left at zero. Lemma 2.1 then preserves feasibility and supplies the interpolant properties needed at the next backward step. The zero terminal table has these properties. Backward induction from $k = N_t - 1$ to $k = 0$ proves the claim. □

The search just analyzed is `solve/search` (lines 135–155); the layer update followed by lowering is in lines 190–201. In particular, no assumption of unimodality is needed for the ternary-search proposals, and no value obtained from an ancestor node is substituted for a literal leaf row.

Finally, the initial table entry is an integer multiple of 2^{-40}, and the reported objective is

$$\gamma = 2^{-40}\lfloor 2^{40}\theta(0) \rfloor + v_{0,N_y,0} \le \theta(0) + v_{0,N_y,0}.$$

The FRLP implication proved in Section 3.3 of the paper, using the concavity of the true V_t and Lemma 3.3, therefore gives

$$L(b, q) \ge \theta(0) + V_0(1, 0) \ge \theta(0) + v_{0,N_y,0} \ge \gamma$$

for every admissible q. The numerical interpolant itself need not be concave. The all-finite-n consequence is supplied by the paper's separate transfer theorem; it is not an additional assertion of this backward evaluator.

3 Certified arithmetic for the frozen evaluator

The numerical analysis applies to the exact source file identified in `PAPER_REFERENCE.json`. The numerical error constants are

$$\varepsilon_{\mathrm{in}} = 2^{-48}, \qquad \varepsilon_{\mathrm{ar}} = 2^{-32} = 65536\varepsilon_{\mathrm{in}}, \qquad \mathfrak{u} = 2^{-53}.$$

Here $\mathfrak{u}$ is the unit roundoff of binary64; the paper's symbol $u = b(1 - y + ty)$ retains its mathematical meaning. The indexed LP coefficients $\eta_{k,i,j,h,i',j'}$ also retain their original meaning; they are unrelated to these numerical error constants.

3.1 Certified evaluation of the continuation terms

This subsection concerns the frozen instance with $N_t = 640$, state knots that are integer multiples of 2^{-30}, and finite rate endpoints at most 8192. It justifies the continuation calculations in `verify.cpp`, including the tests that return a zero continuation. Recall the rounding allowances

$$\varepsilon_{\mathrm{in}} = 2^{-48}, \qquad \varepsilon_{\mathrm{ar}} = 2^{-32} = 65536\varepsilon_{\mathrm{in}}, \qquad \mathfrak{u} = 2^{-53} = \varepsilon_{\mathrm{in}}/32.$$

The code calls the first two quantities `eta` and `eps`. All floating-point operations below are binary64 operations in round-to-nearest mode with gradual underflow (no flush-to-zero or denormals-are-zero mode), without contraction or reassociation. We write fl for the result of one such operation. The usual bound, also valid when its result is subnormal, is

$$|\,\mathrm{fl}(z) - z| \le \mathfrak{u}|z| + 2^{-1075}. \tag{10}$$

None of the operations considered below overflows. The displayed bounds retain ample strict slack for the additive term in (10); in particular, a positive state coordinate satisfies $y \ge 2^{-30}$, and $2^{-1075} < 2^{-990}\varepsilon_{\mathrm{in}} y$.

Exact quantities and validated inputs. Put $\Delta = 1/640$ and $t = k\Delta$, where $0 \le k < 640$. At a nonterminal state, $0 \le x < y \le 1$. The divisions that form x and y from the integer knots and 2^{30} are exact. Write b for the exact drop rate on this time interval and $D = 1 - \theta(t)$. The input validator establishes

$$\tfrac{1}{4} < b < 2, \qquad \tfrac{17}{20} < c < \tfrac{9}{10}, \qquad 0 \le D \le 1.$$

For a finite exact endpoint $r \in [0, 8192]$, define

$$E(r) = e^{-\Delta r}, \quad I_r = \int_0^{\Delta} e^{-rs}\, ds.$$

Thus $0 < E(r) \le 1$, $0 < I_r \le \Delta$, and $0 \le (r + e^{-r} - 1) \le 8192$. The frozen values of $b, c, D, r, E(r), I_r, (r + e^{-r} - 1)$ differ from these exact quantities by less than $\varepsilon_{\mathrm{in}}$ in absolute value. The source entry `vv[h]` is the frozen approximation of $r + e^{-r} - 1$. Both arrays `bs` and `bl` approximate the same exact b. The frozen $\widehat{\Delta}$ satisfies the stronger relative bound

$$|\widehat{\Delta} - \Delta| < \Delta\varepsilon_{\mathrm{in}}.$$

The computed $\widehat{t} = \mathrm{fl}(k\widehat{\Delta})$ consequently satisfies

$$|\widehat{t} - t| \le k\Delta\varepsilon_{\mathrm{in}} + \mathfrak{u}k\Delta(1 + \varepsilon_{\mathrm{in}}) + 2^{-1075} < 2\varepsilon_{\mathrm{in}}. \tag{11}$$

Input validation uses rational enclosures; these input-error assumptions therefore do not assume accuracy of a platform exponential routine.

Recall the exact continuation quantities

$$\begin{aligned}
\varphi(r) &= 1 - y + y\big[t + I_r + E(r)(1 - t - \Delta)\big],\\
y(r) &= \frac{yE(r)}{\varphi(r)}, \qquad \rho(r) = \frac{x}{y} + \frac{b\Delta(r + e^{-r} - 1)}{c},\\
\widetilde{x}(r) &= y(r)\rho(r), \qquad x(r) = \min\{y(r), \widetilde{x}(r)\}.
\end{aligned}$$

We will use

$$E(r) \le \varphi(r) \le 1, \qquad 0 \le y(r) \le 1. \tag{12}$$

For the lower inequality, subtract $E(r)$ from $\varphi(r)$ and use $I_r \ge \Delta E(r)$; for the upper inequality use $I_r \le \Delta$ and $t + \Delta \le 1$.

Elementary operations used by the guards. Here and below an intermediate is evaluated in the order prescribed by the source. Repeated application of (10) gives the following bounds. The table also records sufficient estimates for each operation, so that no unquantified error-propagation assertion is needed. Error bounds in its last column are multiples of $\varepsilon_{\mathrm{in}}$.

Computed intermediate	Exact target	Error bound
$\mathrm{fl}(\widehat{b}\widehat{\Delta})$	$b\Delta$	4Δ
$\mathrm{fl}(\mathrm{fl}(\widehat{b}\widehat{\Delta})\texttt{vv[h]})$	$b\Delta(r + e^{-r} - 1)$	54
$\mathrm{fl}(y\widehat{b})$	yb	$9y/8$
$\mathrm{fl}(\mathrm{fl}(y\widehat{b})\widehat{\Delta})$	$yb\Delta$	$4\Delta y$
$\mathrm{fl}(\mathrm{fl}(\mathrm{fl}(y\widehat{b})\widehat{\Delta})\texttt{vv[h]})$	$yb\Delta(r + e^{-r} - 1)$	$54y$
$\mathrm{fl}(\widehat{c}x)$	cx	$2y$
$\mathrm{fl}(\widehat{D} - \widehat{b})$	$D - b$	3
$\mathrm{fl}(y\,\mathrm{fl}(\widehat{D} - \widehat{b}))$	$y(D - b)$	$4y$

For example, the first line follows from

$$\Delta\varepsilon_{\mathrm{in}} + (2 + \varepsilon_{\mathrm{in}})\Delta\varepsilon_{\mathrm{in}} + \mathfrak{u}(2 + \varepsilon_{\mathrm{in}})\Delta(1 + \varepsilon_{\mathrm{in}}) + 2^{-1075} < 4\Delta\varepsilon_{\mathrm{in}}.$$

For the second line the propagated input error and the new rounding error are at most

$$4\Delta\, 8192\varepsilon_{\mathrm{in}} + 0.004\varepsilon_{\mathrm{in}} + 25.61\mathfrak{u} + 2^{-1075} < 54\varepsilon_{\mathrm{in}}.$$

The same calculation multiplied by y proves the fifth line. The other lines use $|D - b| \le 2$, $c < 0.9$, and $\mathfrak{u} = \varepsilon_{\mathrm{in}}/32$. The final additions in the code therefore give

$$|\widehat{p} - p| < 8\varepsilon_{\mathrm{in}} y, \qquad p = cx + y(D - b), \tag{13}$$

$$|\widehat{n}(r) - n(r)| < 64\varepsilon_{\mathrm{in}} y, \qquad n(r) = cx + yb\Delta(r + e^{-r} - 1), \tag{14}$$

because $|p| \le 3y$ and $0 \le n(r) \le 26.5y$. The right-hand side of the leaf clipping test is $\widehat{n}_{\mathrm{clip}} = \mathrm{fl}(\mathrm{fl}(\widehat{c}y) + \varepsilon_{\mathrm{ar}} y)$. Multiplication by $\varepsilon_{\mathrm{ar}}$ is exact, and

$$|\widehat{n}_{\mathrm{clip}} - (cy + \varepsilon_{\mathrm{ar}} y)| < 2\varepsilon_{\mathrm{in}} y. \tag{15}$$

The factor y in both bounds is essential for small state coordinates.

For the node guard the code computes $\widehat{\rho} = \mathrm{fl}(\mathrm{fl}(x/y) + \mathrm{fl}(\widehat{\omega}/\widehat{c}))$, where $\widehat{\omega}$ is the second intermediate in the table. Since $b\Delta(r + e^{-r} - 1) \le 25.6$, $(b\Delta(r + e^{-r} - 1))/c < 31$, and $\widehat{c} > 0.84$, division has error at most

$$\frac{(54 + 31)\varepsilon_{\mathrm{in}}}{0.84} + 32\mathfrak{u} + 2^{-1075}.$$

Including the division x/y and the final addition yields

$$|\widehat{\rho} - \rho(r)| < 128\varepsilon_{\mathrm{in}}. \tag{16}$$

Likewise, for the product in the upper-endpoint guard,

$$|\,\mathrm{fl}(\widehat{r}\widehat{\Delta}) - r\Delta| < (0.002 + 12.8 + 0.401)\varepsilon_{\mathrm{in}} < 16\varepsilon_{\mathrm{in}}. \tag{17}$$

Validity of the zero-continuation and range tests. If the leaf test skips interpolation, then $\widehat{n}(r) \geq \widehat{n}_{\mathrm{clip}}$. Equations (14) and (15) imply

$$n(r) - cy \geq (\varepsilon_{\mathrm{ar}} - 66\varepsilon_{\mathrm{in}})y > 0.$$

Thus $\rho(r) \geq 1$, the exact successor is on the diagonal, and its continuation is zero. If the leaf test instead evaluates the successor, the reverse estimates give

$$\rho(r) < 1 + \frac{\varepsilon_{\mathrm{ar}} + 66\varepsilon_{\mathrm{in}}}{c} < 1 + 2\varepsilon_{\mathrm{ar}}.$$

For the node test, $\widehat{\rho} \geq 1 + \varepsilon_{\mathrm{ar}}$ implies $\rho(r) > 1$ by (16), so all its descendant continuations are zero by the monotonicity of ρ. Otherwise $\rho(r) < 1 + 2\varepsilon_{\mathrm{ar}}$ again. In every case where a lower-endpoint successor is evaluated, $(r + e^{-r} - 1) \geq r - 1$ now implies

$$r\Delta \leq \Delta + \frac{c}{b}\rho(r) < \frac{1}{640} + 3.6(1 + 2\varepsilon_{\mathrm{ar}}) < 4. \tag{18}$$

For an upper endpoint, the code evaluates its successor only if $\mathrm{fl}(\widehat{r}\widehat{\Delta}) \leq 4 - \varepsilon_{\mathrm{ar}}$. Equation (17) proves $r\Delta < 4$ in this case as well. Otherwise the node calculation uses the valid lower bound zero for that successor's y-coordinate.

Errors in φ and the successor coordinates. The following table follows the parenthesization of `ph`. Its error bounds are again in units of $\varepsilon_{\mathrm{in}}$.

Computed intermediate	Exact target	Error bound
$\mathrm{fl}(\mathrm{fl}(1 - \widehat{t}) - \widehat{\Delta})$	$1 - t - \Delta$	3
$\mathrm{fl}(\widehat{E}\widehat{K})$	$E(1 - t - \Delta)$	5
$\mathrm{fl}(\widehat{t} + \widehat{I})$	$t + I_r$	4
$\mathrm{fl}(\widehat{T} + \widehat{E}\widehat{K})$	$t + I_r + E(1 - t - \Delta)$	10
$\mathrm{fl}(y\widehat{Z})$	$y[t + I_r + E(1 - t - \Delta)]$	$11y$

In the fourth line $\widehat{E}\widehat{K}$ denotes the already rounded second-line product. Here $\widehat{K}, \widehat{T}, \widehat{Z}$ denote the corresponding previously computed quantities. For completeness, the first-line estimate is bounded by $2 + \Delta + (1 + \Delta)/32$ in the stated units, with an arbitrarily small allowance for $\varepsilon_{\mathrm{in}}$ and the subnormal term; the next lines are bounded respectively by $3 + 1 + 1/32$, $2 + 1 + 1/32$, $5 + 4 + 1/32$, and $10y + y/32$, with the same strict slack. The quantity $1 - y$ is an exactly representable multiple of 2^{-30}. The last addition thus gives

$$|\widehat{\varphi}(r) - \varphi(r)| < 16\varepsilon_{\mathrm{in}}. \tag{19}$$

For every evaluated successor, (18) and (12) give $\varphi(r) > e^{-4} > 1/64$, hence $\widehat{\varphi}(r) > 1/65$. The rounded numerator $\widehat{a} = \mathrm{fl}(y\widehat{E})$ satisfies $|\widehat{a} - yE| < 2\varepsilon_{\mathrm{in}}y$. Using $y(r) \leq 1$, the error before rounding the division is at most

$$\left|\frac{\widehat{a}}{\widehat{\varphi}} - y(r)\right| \leq \frac{|\widehat{a} - yE| + y(r)|\widehat{\varphi} - \varphi|}{\widehat{\varphi}} < 65(2y + 16)\varepsilon_{\mathrm{in}}.$$

The division adds less than $2\mathfrak{u} + 2^{-1075}$, proving

$$|\widehat{y}(r) - y(r)| < 2048\varepsilon_{\mathrm{in}}. \tag{20}$$

The leaf computes an *unclipped* x-coordinate by dividing $\mathrm{fl}(\widehat{E}\widehat{n})$ by $\mathrm{fl}(\widehat{c}\widehat{\varphi})$. Under its evaluation guard, $n = cy\rho < y$ and $\widetilde{x}(r) \leq 1 + 2\varepsilon_{\mathrm{ar}} < 1.001$. Equation (14) and one multiplication give numerator error less than $66\varepsilon_{\mathrm{in}}y$. Equation (19) gives denominator error

less than $16\varepsilon_{\text{in}}$, and the computed denominator is greater than $1/80$. The same quotient identity therefore bounds the error before the final rounding by

$$80(66y + 16 \cdot 1.001)\varepsilon_{\text{in}}.$$

Including the division rounding yields

$$|\widehat{x}(r) - \widetilde{x}(r)| < 8192\varepsilon_{\text{in}}. \tag{21}$$

For a finite upper endpoint that passes its range guard, the node instead computes $\widehat{x}_{\text{low}} = \text{fl}(\widehat{y}(r_+)\min\{1, \widehat{\rho}(r_-)\})$. The map $z \mapsto \min\{1, z\}$ is 1-Lipschitz. Combining (16) and (20), then allowing one product rounding, gives

$$\left|\widehat{x}_{\text{low}} - y(r_+)\min\{1, \rho(r_-)\}\right| < 4096\varepsilon_{\text{in}}. \tag{22}$$

For example, the coefficient of ε_{in} is bounded by $2048(1 + 128\varepsilon_{\text{in}}) + 128 + 2/32$, which is below 4096. When the upper-endpoint guard is not used, the assigned value is exactly zero and needs no floating-point estimate.

Shifted interpolation and multiplication. Let $\widetilde{v}_{k+1}$ be the ordinary triangular interpolant already defined above; its vertex values are stored integers divided by 2^{40}. The monotonicity passes ensure $\widetilde{v}_{k+1} \le 0$, that $\widetilde{v}_{k+1}$ is nonincreasing in its first coordinate and nondecreasing in its second coordinate, and that it is zero on the diagonal. The interpolation routine rounds $\widehat{y} + \varepsilon_{\text{ar}}$ up to a multiple of 2^{-30} and $\widehat{x} - \varepsilon_{\text{ar}}$ down to such a multiple, then clips the former to $[0, 1]$ and the latter below at zero. The additions incur less than $\varepsilon_{\text{in}}/8$ error; multiplication by 2^{30} is exact. Equations (20)– (22) and $\varepsilon_{\text{ar}} = 65536\varepsilon_{\text{in}}$ therefore put the integer query above the exact required y-coordinate and below the exact required x-coordinate. Clipping at 0 or 1 preserves these inequalities.

For a leaf with $\rho(r) < 1$, the required x-coordinate is $\widetilde{x}(r) = x(r) \le y(r)$, so the queried value is at most $\widetilde{v}_{k+1}(y(r), x(r))$. If the routine returns zero because its integer x-coordinate is no smaller than its integer y-coordinate, these coordinate inequalities force the exact successor to be on the diagonal; the zero return is then valid. For a leaf with $\rho(r) \ge 1$, the true continuation is already zero. In this case either a zero return or any nonpositive interpolated value is conservative, even though the code formed the raw coordinate $\widetilde{x}(r)$ rather than its clipped version. The identical argument applies to a node when $\rho(r_-) < 1$ and the required coordinate is $y(r_+)\rho(r_-)$; when $\rho(r_-) \ge 1$, every descendant continuation is zero. These are exactly the cases used by the exact-arithmetic node bound.

Inside a containing triangle, the code forms nonnegative integer barycentric numerators with their exact sum equal to the positive denominator, and takes a downward integer quotient. Thus the returned integer v satisfies $v/2^{40} \le \widetilde{v}_{k+1}$ at the integer query. Integer range bounds are established separately below.

Finally, the code uses

$$M = \left\lceil \text{fl}\big((\widehat{\varphi} + \varepsilon_{\text{ar}})2^{40}\big) \right\rceil + 1,$$

where the parenthesized addition is rounded once and multiplication by 2^{40} is exact. Its addition error is less than $\varepsilon_{\text{in}}/8$. Consequently

$$\varphi \le \frac{M}{2^{40}} \le \varphi + 2\varepsilon_{\text{ar}} + 2 \cdot 2^{-40}.$$

Since $v \le 0$, the exact downward quotient used for $Mv/2^{40}$ certifies a lower bound on the exact multiplier times the required interpolant. This proves the rounding correctness of both the leaf continuation and the subtree continuation bound used for pruning.

3.2 Certification of the running-cost arithmetic

This subsection concerns the frozen functions `B`, `Bnode`, and `log_lower`; it does not introduce a different running-cost bound. We use the constants $\varepsilon_{\mathrm{in}} = 2^{-48}$, $\varepsilon_{\mathrm{ar}} = 2^{-32}$, and $\mathfrak{u} = 2^{-53}$ fixed above. Write $\mathfrak{g}_j = j\mathfrak{u}/(1-j\mathfrak{u})$. Hats denote computed binary64 quantities. In particular, $|\widehat{p} - p| < 8\varepsilon_{\mathrm{in}}$ and $|\widehat{t} - t| < 2\varepsilon_{\mathrm{in}}$, as established in the continuation analysis. The validated data satisfy

$$\Delta = 1/640, \quad 0 \le t \le 1-\Delta, \quad 0 \le x < y \le 1,$$
$$\tfrac{1}{4} < b < 2, \quad |p| < 2, \quad 0 \le I_r \le \Delta, \quad 0 \le E(r), e^{-r} \le 1.$$

Here $p = cx + y(D-b)$, where $.85 < c < .9$ and $0 \le D \le 1$: thus $p > -2$ and $p \le y(.9 + 1 - 1/4) \le 1.65 < 2$. Each stored coefficient has absolute error less than $\varepsilon_{\mathrm{in}}$, and the stored Δ has relative error less than $\varepsilon_{\mathrm{in}}$. Write

$$p_- = -(-p)^+, \qquad p_+ = p^+, \qquad C(r) = p_-\Delta + p_+ I_r,$$

$$\alpha(r) = \left(ybE(r) - \frac{p_- I_r}{\Delta}\right)\Delta^2, \quad \beta(r) = b(1-y+ty)\Delta + ybI_r\Delta, \quad z(r) = \beta(r)e^{-r}.$$

A reproducible error bound for the coefficients. For every elementary operation, including one with a subnormal result, we use the following weaker form of the preceding floating-point model:

$$|\operatorname{fl}(v) - v| \le \mathfrak{u}|v| + 2^{-1074}. \tag{23}$$

For example, if $|a| \le A$, $|b| \le B$ and the operand errors are e_a, e_b, a product has error at most

$$Ae_b + Be_a + e_a e_b + \mathfrak{u}(A + e_a)(B + e_b) + 2^{-1074}.$$

An addition has error at most $e_a + e_b + \mathfrak{u}(A + B + e_a + e_b) + 2^{-1074}$. For division by a positive exact $b \ge b_0 > e_b$, first bound the input perturbation by $(e_a + |a/b|e_b)/(b_0 - e_b)$, and then apply (23). The maps $v \mapsto \min(v, 0)$ and $v \mapsto \max(v, 0)$ do not increase absolute errors.

The following table applies these inequalities in the source's evaluation order. The middle column bounds the magnitude of the corresponding exact expression, and the last column is a strict error bound in units of $\varepsilon_{\mathrm{in}}$. Each multiplication by Δ is a separate operation. The dyadic quantity $1 - y$ is exact.

Expression	Exact magnitude bound	Error coefficient
yb; ybE	2; 2	2; 5
p_-I_r; p_-I_r/Δ	2Δ; 2	3; 2000
$ybE - p_-I_r/\Delta$	4	2010
$(ybE - p_-I_r/\Delta)\Delta$	4Δ	4
$\alpha = (ybE - p_-I_r/\Delta)\Delta^2$	$4\Delta^2$	1
ty; $1 - y + ty$	1; 1	3; 4
$b(1-y+ty)$; $b(1-y+ty)\Delta$	2; 2Δ	10; 1
ybI_r; $ybI_r\Delta$	2Δ; $2\Delta^2$	3; 1
β; $z = \beta e^{-r}$	$2\Delta(1+\Delta) < .004$	3; 4
$p_-\Delta$; p_+I_r; $C = p_-\Delta + p_+I_r$	2Δ	1; 3; 5

For the largest intermediate error, for example, division of the preceding $3\varepsilon_{\mathrm{in}}$ error by $\Delta(1-\varepsilon_{\mathrm{in}})$ contributes less than $1921\varepsilon_{\mathrm{in}}$; denominator perturbation and rounding contribute less than $3\varepsilon_{\mathrm{in}}$. The two following multiplications reduce the bound to the displayed 4 and 1. All other entries follow by the same displayed product and sum rules; the strict margins also exceed the sum of the fewer than 100 possible 2^{-1074} terms. In particular, all four final errors are less than $8\varepsilon_{\mathrm{in}}$.

The logarithm computed by the frozen source. This paragraph certifies the twelve-term logarithm loop in the source. Whenever the logarithmic branch is reached, put $v = \widehat{\beta}/\widehat{\alpha}$ in exact real arithmetic and $\widehat{v} = \mathrm{fl}(v)$. The guards give $\widehat{\beta} > \widehat{\alpha} > \varepsilon_{\mathrm{ar}}$; hence $1 \le \widehat{v} < 2^{25}$ and $|\log \widehat{v} - \log v| < 2\mathfrak{u}$. The exact binary operation `frexp` gives $\widehat{v} = 2^e m$ with $0 \le e \le 24$ and $1 \le m < 2$. For $w_0 = (m-1)/(m+1)$,

$$S_{12}(m) = 2\sum_{j=0}^{11} \frac{w_0^{2j+1}}{2j+1} \le \log m, \qquad 0 \le w_0 < \tfrac{1}{3}.$$

The subtraction $m-1$ is exact. Rounding of $m+1$ and the division give a relative error bounded by $\mathfrak{g}_2$ in a nonzero w_0. In the largest monomial, its degree is 23; the repeated squaring-factor and product operations contribute at most 22 further factors, division contributes one, and summation contributes at most eleven. Thus the computed $2\,$`sum` is at most

$$(1 + \mathfrak{g}_{128}) S_{12}(m) + 256 \cdot 2^{-1074}.$$

This bound also covers underflow of the smallest positive monomials. There is no assumption about a platform logarithm implementation.

The hexadecimal constant in the source is the exact rational

$$c_2 = \frac{3121657384082679}{2^{52}} = \texttt{0x1.62e42fefa39eep-1}.$$

The finite rational inequality

$$2\sum_{j=0}^{39} \frac{1}{(2j+1)3^{2j+1}} > c_2 + 2^{-53}$$

is checked by multiplication by the positive common denominator. The left side is a lower partial sum of the positive series for $\log 2$, so $c_2 < \log 2$. Since $S_{12}(m) < 1$ and $e \le 24$, the above polynomial error, the product ec_2, the final addition and subtraction, and the rounded quotient have combined upper error at most

$$\mathfrak{g}_{128} S_{12}(m) + 24\mathfrak{u} + 25\mathfrak{u} + 26\mathfrak{u} + 2\mathfrak{u} + 512 \cdot 2^{-1074} < 256\mathfrak{u} + 512 \cdot 2^{-1074} < 32\varepsilon_{\mathrm{in}}.$$

Consequently the returned value satisfies

$$\texttt{log_lower}(\widehat{v}) \le \log v - \varepsilon_{\mathrm{ar}} + 32\varepsilon_{\mathrm{in}} < \log v. \tag{24}$$

The free bound and its branch tests. Recall $\mathcal{H}$ from (4). On the guarded domain and on the segment joining the exact and computed coefficient pairs, $\alpha > \varepsilon_{\mathrm{ar}}/2$ and $\beta < .004$. The function is continuously differentiable across $\beta = \alpha$, and direct differentiation gives

$$0 \le \partial_\beta \mathcal{H} \le 1, \qquad 0 \le \partial_\alpha \mathcal{H} < 10.$$

Indeed $\beta/\alpha < 2^{26}$ in the logarithmic region. Thus coefficient perturbations change $\mathcal{H}$ by less than $88\varepsilon_{\mathrm{in}}$. This argument permits the computed and exact pairs to fall on different sides of the branch test. In the rational branch the division, subtraction, and multiplication contribute less than $\varepsilon_{\mathrm{in}}$ in absolute error. In the other branch, (24) and the remaining two rounded operations give the same one-sided bound. Including the error of `base`, addition, and subtraction, the computed candidate `base+freebound-eps` is therefore at most

$$C(r) + \mathcal{H}(\alpha(r), \beta(r)) - \varepsilon_{\mathrm{ar}} + 128\varepsilon_{\mathrm{in}},$$

and is a valid lower bound. If a positivity guard fails, the source simply does not use this candidate. The baseline `min(p,0)*dt-eps` is separately a lower bound for $\min(p, 0)\Delta$.

The scalar shifted bound. The frozen rate grid contains 64 exactly and every other rate has distance at least 16 from 64. Thus `rr[hi]<=64` cannot admit an exact rate exceeding 64. The test $\widehat{z} > \mathrm{fl}(\widehat{\alpha} + \varepsilon_{\mathrm{ar}})$ implies $z > \alpha$: the total coefficient and addition errors are less than $17\varepsilon_{\mathrm{in}}$. The same strict inequality holds throughout the segment between the exact and computed (α, z) pairs. For

$$\mathcal{J}(\alpha, z, r) = z + \alpha r/2 - \frac{\alpha^2}{24(z - \alpha/2)}, \qquad z > \alpha \geq 0,$$

differentiation gives

$$1 \leq \partial_z \mathcal{J} \leq \tfrac{7}{6}, \qquad |\partial_\alpha \mathcal{J}| \leq 33, \qquad |\partial_r \mathcal{J}| = \alpha/2 < 1 \quad (0 \leq r \leq 64 + \varepsilon_{\mathrm{in}}).$$

Hence coefficient perturbations, including that of r, contribute less than $275\varepsilon_{\mathrm{in}}$. The subtraction in $\ell = z - \alpha/2$ has no harmful relative cancellation, since $\ell > z/2$ and $\ell > \alpha/2$. The exact quotient $\alpha^2/(24\ell)$ is at most $\alpha/12$. The rounding of ℓ, its multiplication by 24, the numerator product, and division give absolute error at most $\mathfrak{g}_5\alpha/12 + 2^{-1000}$ in this quotient. For the additive term, the guard gives $\ell > \varepsilon_{\mathrm{ar}}/2$; propagating the fewer than ten possible 2^{-1074} errors through these operations amplifies them by less than 2^{40}, which is covered by 2^{-1000}. All remaining products and sums have magnitude less than 1; applying (23) to them and to `base` bounds the complete upward error by $512\varepsilon_{\mathrm{in}}$. Thus `sharp-eps` is below the exact shifted expression whenever this branch is used.

The shifted bound for a tree node. Let the node's endpoint rates be $r_- \leq r_+ \leq 64$. The quantities actually formed by the source are

$$\overline{\alpha} = \mathrm{fl}(\widehat{\alpha}(r_-) + 16\varepsilon_{\mathrm{in}}), \quad \underline{\alpha} = \max\{0, \mathrm{fl}(\widehat{\alpha}(r_+) - 16\varepsilon_{\mathrm{in}})\},$$

$$\underline{z} = \max\{0, \mathrm{fl}(\widehat{z}(r_+) - 16\varepsilon_{\mathrm{in}})\}, \quad \underline{r} = \max\{0, \mathrm{fl}(\widehat{r}_- - 16\varepsilon_{\mathrm{in}})\}.$$

The table above proves $\overline{\alpha} \geq \alpha(r_-)$, $0 \leq \underline{\alpha} \leq \alpha(r_+)$, and $0 \leq \underline{z} \leq z(r_+)$: the extra additions and subtractions have error less than $\varepsilon_{\mathrm{in}}$. For the last expression their error is at most $3\varepsilon_{\mathrm{in}}$, so $0 \leq \underline{r} \leq r_-$ as well. The rounded guard implies $\underline{z} > \overline{\alpha}$. Therefore the exact expression in these already directed floating inputs,

$$C(r_+) + \underline{z} + \underline{\alpha}\,\underline{r}/2 - \frac{\overline{\alpha}^2}{24(\underline{z} - \overline{\alpha}/2)},$$

is a lower bound for the shifted cost of every descendant original row, by the endpoint argument proved above. Its quotient is again at most $\overline{\alpha}/12$, with absolute operation error at most $\mathfrak{g}_5\overline{\alpha}/12 + 2^{-1000}$. The remaining operation errors and the error of `base` together are less than $16\varepsilon_{\mathrm{in}}$. Subtracting $2\varepsilon_{\mathrm{ar}}$ therefore preserves the lower-bound direction. No monotonicity of the full shifted expression is assumed.

In `B`, the maximum of the available certified candidates remains certified. The final subtraction of $\varepsilon_{\mathrm{ar}}$, exact power-of-two scaling, floor, and subtraction of one integer unit give a lower bound in units of 2^{-40}. The analogous floor and subtraction in `Bnode` have the same direction. The magnitudes just established are below 1, so the final floating subtraction has error less than $\varepsilon_{\mathrm{in}}$ and cannot exhaust any of these margins.

3.3 Table magnitudes, integer safety, and the final objective

The allowance computation requires no approximation theorem. The coordinates are exact multiples of 2^{-30}, so its $w = y - x$ is exact and lies in $[0, 1]$. Whatever error occurs in computing

the intermediate `future`, the two comparisons in `max(0,min(w,w-future))` put the returned floating value in $[0, w]$. Multiplication by 2^{24} is exact, and consequently

$$0 \le \texttt{allowance} \le 2^{24}.$$

Subtracting this integer therefore only lowers a proposed table value, by at most 2^{-16}. The fixed additional slack is 2^{-28}.

Let M_k be the largest absolute table entry in layer k, after division by 2^{40}. All entries are nonpositive. Integer barycentric interpolation of the next layer lies between its minimum and zero: flooring cannot cross the integer minimum. The previously certified multiplier, including its ceiling and extra integer unit, is at most $2^{40}(1 + 2\varepsilon_{\mathrm{ar}} + 2 \cdot 2^{-40})$. The subsequent integer floor loses less than 2^{-40}. For the running cost, the actual baseline arithmetic gives

$$2^{-40}\texttt{B(hi)} \ge -3/640 - 3\varepsilon_{\mathrm{ar}} - 2 \cdot 2^{-40}.$$

Indeed $|\widehat{p}| < 2 + 8\varepsilon_{\mathrm{in}}$; the computed product with $\widehat{\Delta}$ is greater than $-3/640$; the baseline and final floating subtractions use two $\varepsilon_{\mathrm{ar}}$ deductions, their rounding is covered by the third, and flooring followed by subtraction of one costs less than two integer units. Taking a maximum can only improve this lower bound. It follows, with deliberately enlarged constants, that

$$\begin{aligned} M_k &\le aM_{k+1} + d, \qquad M_{640} = 0, \\ a &= 1 + 2\varepsilon_{\mathrm{ar}} + 2 \cdot 2^{-40}, \\ d &= 3/640 + 4\varepsilon_{\mathrm{ar}} + 8 \cdot 2^{-40} + 2^{-28} + 2^{-16}. \end{aligned}$$

Taking minima with zero or with additional proposals, and either monotonicity-lowering pass, preserves this global lower bound. Since $640(a-1) < 1$, the binomial theorem gives

$$M_k \le d \sum_{j=0}^{639} a^j \le \frac{640d}{1 - 640(a-1)} = \frac{1846693964}{613566573} < \frac{301}{100} < 4.$$

Thus every stored integer has magnitude below $4 \cdot 2^{40}$.

Whenever interpolation is performed, its tests give $0 \le X < Y \le 2^{30}$. Lookup indices lie in $[0, 65536]$, and every grid width and offset lies in $[0, 2^{30}]$. The barycentric denominator and each unscaled weight are at most 2^{60}, so their products and differences fit signed 64-bit integers. The selected triangle has nonnegative weights summing to the denominator. Its weighted numerator and all partial sums therefore have magnitude at most $4 \cdot 2^{40} \cdot 2^{60} = 2^{102}$, safely within signed 128-bit range. Continuation multiplier products are smaller than 2^{84}, and the negative-floor formula's additional denominator is also harmless at these bounds. Each conversion of a running-cost value, multiplier, or allowance to a signed 64-bit integer is within its range. The state-array size is $1201 \cdot 1202/2 = 721801$; all index products fit signed 32-bit integers. A binary tree over 131093 leaves has fewer than twice that many nodes; the additional proposal evaluations are fewer than 50 per state. Per-state counters fit unsigned 32-bit integers and their sums over all states fit unsigned 64-bit integers.

Finally, input validation establishes $\texttt{HU} = \lfloor 2^{40}\theta(0) \rfloor$. The reported exact objective is therefore

$$\frac{\texttt{HU} + \texttt{table}[m, 0]}{2^{40}} \le \theta(0) + \frac{\texttt{table}[m, 0]}{2^{40}}.$$

Together with the original-row feasibility induction, this proves the claimed direction of the rational objective; the printed decimal plays no role in acceptance.

4 Source mapping and reproduction

The source lines cited in this document refer to the unchanged `certificate/verify.cpp`. Its SHA-256 is

`6cafb1fb9e1be830c8b2c8cd60304bb1e2c3496e3da5a3e7e2d47cb4b9a84219`.

The release preflight checks this identity and all 41 entries in the current certificate manifest. In particular, the mathematical argument is tied to the actual expression order and guards of this source.

Source item	Paper notation	Role
`N`, `Q`, `m`	N_t, N_q, N_y	Time, rate, and state grid dimensions
`bs[k]`, `bl[k]`	b on $[t_k, t_{k+1}]$	Identical approximations of the same exact rational drop rate
`DD[k]`	$1 - \theta(t_k)$	Frozen threshold complement
`cap`	c	The constant from Appendix A.1
`ee[h]`, `ii[h]`	$e^{-\Delta r_h}, I_{r_h}$	Rate-dependent input coefficients
`vv[h]`, `er[h]`	$r_h + e^{-r_h} - 1, e^{-r_h}$	Rate-dependent input coefficients
`A`, lines 63–68	Continuation in (5)	Leaf continuation with conservative coordinates and multiplier
`B`, lines 78–96	Lemma A.3 cost	Conservative evaluation of available cost bounds
`rectA`, lines 98–106	Lemma 2.2	Continuation bound for a whole rate subtree
`Bnode`, lines 108–124	Lemmas 2.3–2.4	Cost bound for a whole rate subtree
`search/solve`, lines 135–155	Theorem 2.5	Complete constraint coverage and downward candidate updates
`lower_monotone`, lines 158–166	Lemma 2.1	Current-layer lowering and explicit monotonicity checks

The scalar storage constant named `U` in the source is 2^{40}, and `L` is the state-grid denominator 2^{30}. These code identifiers do not redefine the paper's mathematical symbols.

4.1 Input validation

The input validator represents every hexadecimal input exactly as a rational number. For $0 \le z \le 1$, the alternating Taylor sums of degrees 25 and 24 enclose e^{-z}. For larger arguments below 64, the validator repeatedly halves the argument until it lies in $[0, 1]$ and squares the positive rational endpoints to recover the original exponential. For $z \ge 64$, it uses

$$0 \le e^{-z} \le 2^{-\lfloor z \rfloor},$$

which follows from $e > 2$. These enclosures validate the entire frozen coefficient file, including the differences used to form I_r and $r + e^{-r} - 1$, with error strictly below ε_{in}. The exact $r = 0$ value is $I_0 = \Delta$. The extra preflight checks establish the fixed-instance facts used by the arithmetic proof and the consistency of all input representations.

4.2 Commands and acceptance conditions

From the release root, run

```
python3 reproduce.py --check-only
python3 reproduce.py --full --threads 8
```

The first command checks the current certificate manifest, the exact preflight assumptions, and all coefficient input enclosures. The second also invokes the launcher, reconstructs all 640 backward layers, and checks the exact objective against `certificate/RESULTS.json`. It compares the complete final current table, next-layer table, and hint array with the original checkpoint hashes. Acceptance never uses a decimal display.

The launcher requires Python 3.9 or later and a GNU-compatible C++17 compiler with OpenMP and signed `__int128`. The compilation options include `-ffp-contract=off`; fast-math and disabled assertions are rejected. The evaluator sets and checks round-to-nearest on each worker before its state computations. Python assertions must also remain enabled; the reproduction script, launcher, coefficient validator, and row checkers reject disabled Python assertions. Standard IEEE gradual underflow is required, with flush-to-zero and denormals-are-zero modes disabled.

The files in `validation/` identify the checks performed. The dated complete replay record and layer log concern all 640 layers of the unchanged evaluator and inputs; current packaging checks are recorded separately. Selected-state diagnostic scans and exact interval row checks are additional, explicitly scoped evidence. Their role is to cross-check the implementation; the full feasibility argument is the induction and arithmetic proof above.

The source, rational inputs, and proof suffice to verify the fixed threshold. No property of the historical threshold-search procedure is used. To rebuild this PDF, run `latexmk -pdf verification.tex` in `docs/`; it repeats LaTeX until references and the table of contents stabilize. The distributed PDF is already compiled.